

# ECE462L Lab Report 2

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# 1 Lab 3: Inductor Design

For Lab 3, our group designed the inductor that would be used for our boost converter. Main aspects to consider in this design were the mosfets, core size, and the wire gauge, which were selected from calculations done in Homework 5 and work done in the Lab. We tried to maximize our efficiency at the expense of size, which is the driving factor behind choosing our values that we plugged into the Kg method equations later on.

## 1.1 MOSFET

To calculate optimal resistance, a switching frequency of 50 kHz and ripple ratio of 0.125 were used. The optimal resistance is:

$$R_{\text{opt}} = \frac{V}{I} \sqrt{\frac{2\tau f_s}{1 + \frac{\text{ripple\_ratio}^2}{3}}} \quad (1)$$

The calculated  $R_{\text{opt}}$  was 2 m $\Omega$ . The IFRB7534 MOSFET was selected, with  $R_{\text{on}} = 2$  m $\Omega$ . The figure of merit  $\tau$  was 1784 fs.

## 1.2 Target Inductance

$$L = \frac{v_{\text{in}}(v_{\text{out}} - v_{\text{in}})}{2f \cdot \text{ripple} \cdot v_{\text{out}} \cdot i} \times 10^6 \quad (2)$$

Using  $V_{\text{in}} = 24\text{V}$ ,  $V_{\text{out}} = 48\text{V}$ , ripple = 0.125, and  $I = 10.417\text{A}$ , the target inductance is 92.16  $\mu\text{H}$ .

### 1.3 Inductor Parameters

Listing 1: MATLAB code to select inductor core

```
1 fprintf("\nCHOOSING AN INDUCTOR CORE\n")
2 for i = 1:length(core_names)
3     compare_value(i) = (rho * ((L*1e-6)^2) * (Imax^2)) / ...
4     ((Bmax(i)^2) * winding_resistance * Ku) * 1e8; % cm^5
5
6     Kg_value(i) = (Ac(i)^2 * WA(i)) / MLT(i);
7
8     if Kg_value(i) > compare_value(i)
9         fprintf("%s works! (Kg = %.3f, required = %.3f)\n", ...
10             core_names{i}, Kg_value(i), compare_value(i));
11         nbmax = Bmax(i);
12         nac = Ac(i);
13         nwa = WA(i);
14         nmlt = MLT(i);
15         core = core_names{i};
16     else
17         fprintf("%s will NOT work. (Kg = %.3f, required = %.3f)\n", ...
18             core_names{i}, Kg_value(i), compare_value(i));
19     end
20 end
```

ETD44 was selected because its  $K_g = 0.8089$  exceeded the required 0.7589.

### 1.4 Air Gap Length

$$l_g = \frac{\mu(L \cdot 10^{-6})I_{\max}^2 \cdot 10^4}{n_{b\max}^2 A_C} \quad (3)$$

Air gap length: 0.00079526 m

### 1.5 Turns Calculation

$$n = \frac{(L \cdot 10^{-6})I_{\max} \cdot 10^4}{n_{b\max} A_C} \quad (4)$$

Turns required: 19

## 1.6 Wire Gauge Selection

Listing 2: MATLAB code to select wire gauge

```
1 for j = 1:length(awg)
2     if nws >= bare_area(j)
3         fprintf("Recommended AWG:\n%d\n", awg(j));
4         Aw = bare_area(j);
5         Gauge = awg(j);
6         break;
7     end
8 end
```

Calculated: 10 AWG Available in lab: 14 AWG used, due to unavailability of the 10 AWG.

## 1.7 LCR Screenshot

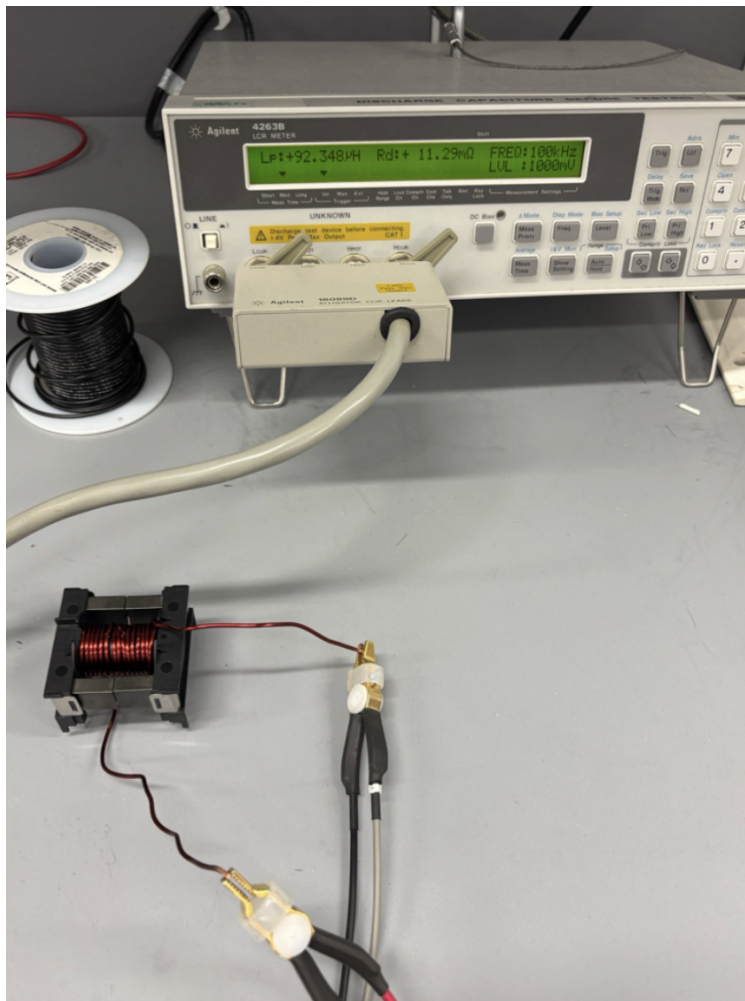


Figure 1: Measured inductance on LCR meter

## 1.8 Actual vs Target Inductance

Measured: 92.348  $\mu\text{H}$  Target: 92.16  $\mu\text{H}$

Difference: 0.2%

Possible error sources: imperfect winding, extra lead length, overlapping turns.

## 2 Lab 4: Gate Driver Testing

### 2.1 Testing Process

PWMA and PWMB signals were checked first to verify deadtimes. Then bootstrap,  $V_{sw}$ , and gate driver outputs were tested using the test points of the PCB. Everytime data was recorded, we ran the code to observe the waveforms of the PWMs, the gate driver, and the bootstrap voltage. Challenges we came across this process is the current sensor was orientated the wrong way, so that hindered our progression a bit. To fix this, we simply just turned the current sensor around.

### 2.2 PWM Screenshots

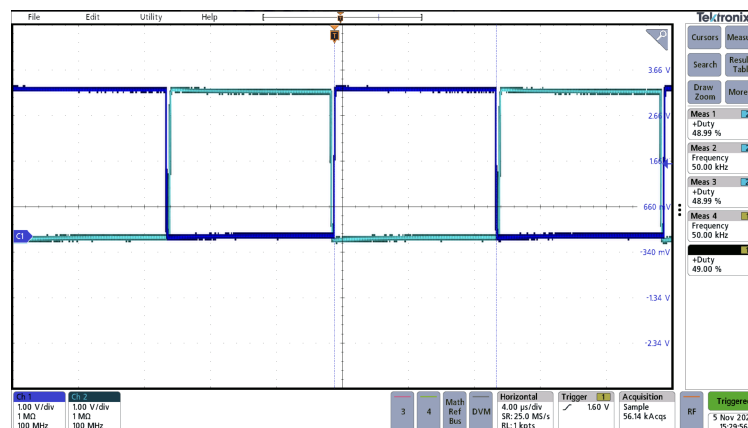


Figure 2: VPWMA and VPWMB signals

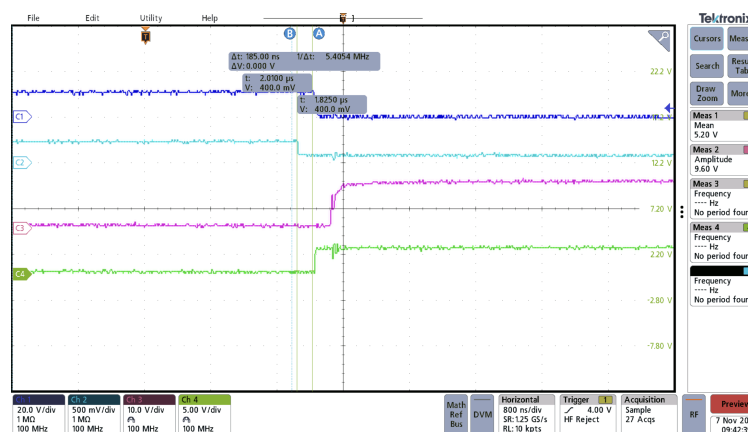


Figure 3: Gate driver deadtime

## 2.3 Waveforms

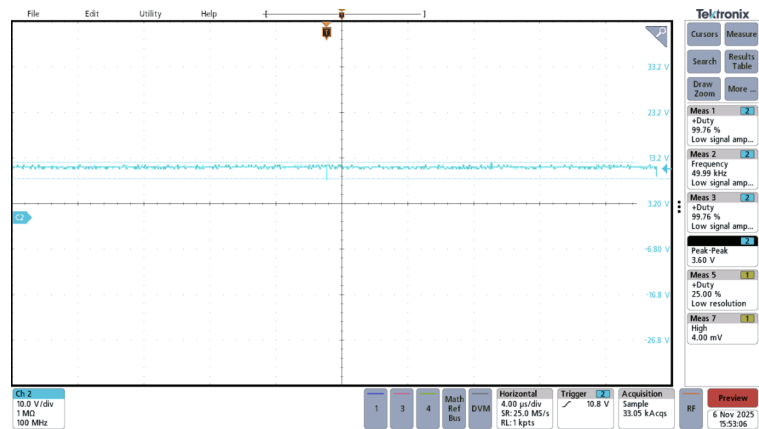


Figure 4: Bootstrap Voltage

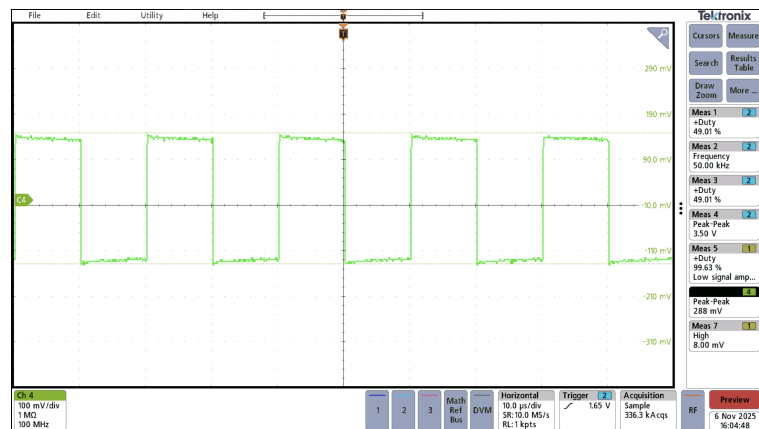


Figure 5: Switch Node Waveform

## 3 Boost Under Load

### 3.1 Testing Process

To start testing our boost converter with actual loads, we began by connecting the smallest load to verify our power stage was still working correctly. While we set up everything correctly, our first test did not run correctly because our MOSFETs were shorted together. This was because we tried to put them as close as possible to reduce the loop area, which caused the screws on the back to touch. By bending the MOSFETs slightly, we fixed this short and successfully tested our converter.

After verifying that the output voltage was reaching around 48 volts, and the waveforms looked reasonable and stable, we would scale up our load. This process repeated until we successfully ran our converter at max load. For our final competition measurements, we ran our converter many times at max load in order to try to increase our final efficiency. During this period of testing, we broke one of our MOSFETs, which shorted our 24V zone to Ground. To debug this, we desoldered all of the components that connect to either Vin or 24V on our board, starting with the DC-DC Converters and ending with the MOSFETs. Luckily, by replacing the MOSFET that broke, we were able to get our converter running again.

### 3.2 Screenshots

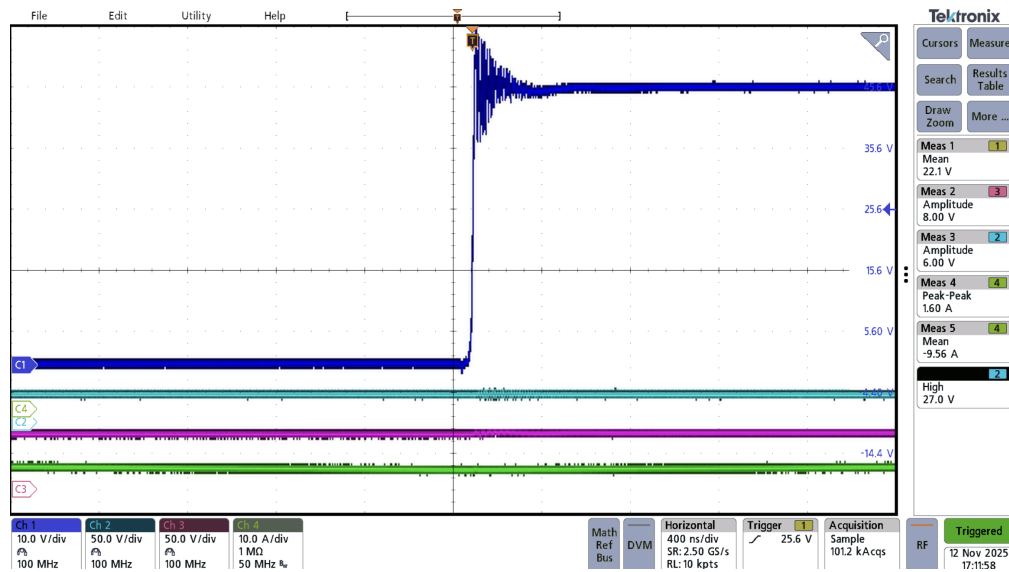


Figure 6: Overshoot of Switch Node Voltage  $V_{sw}$

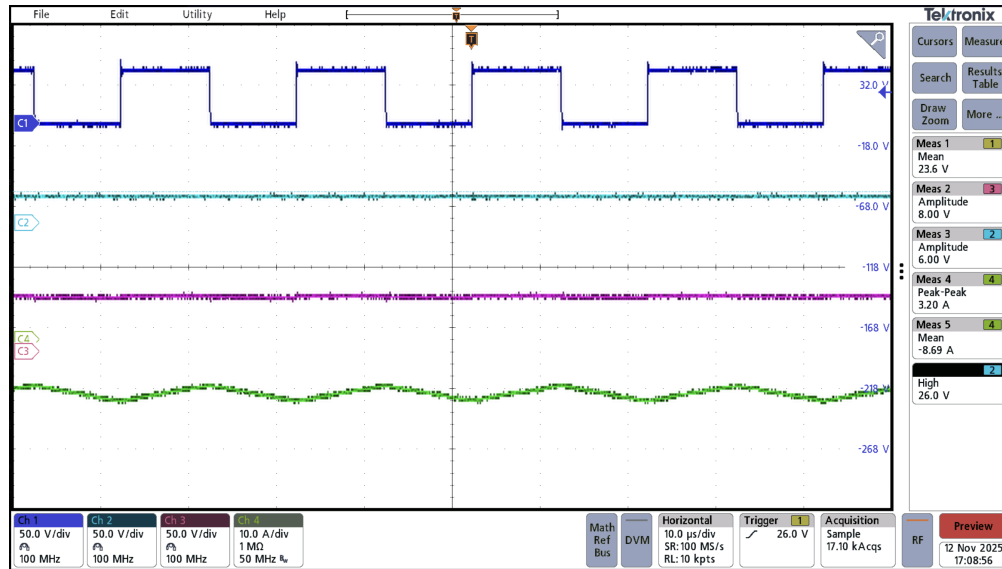


Figure 7: Inductor Current  $I_L$  and Ripple Ratio

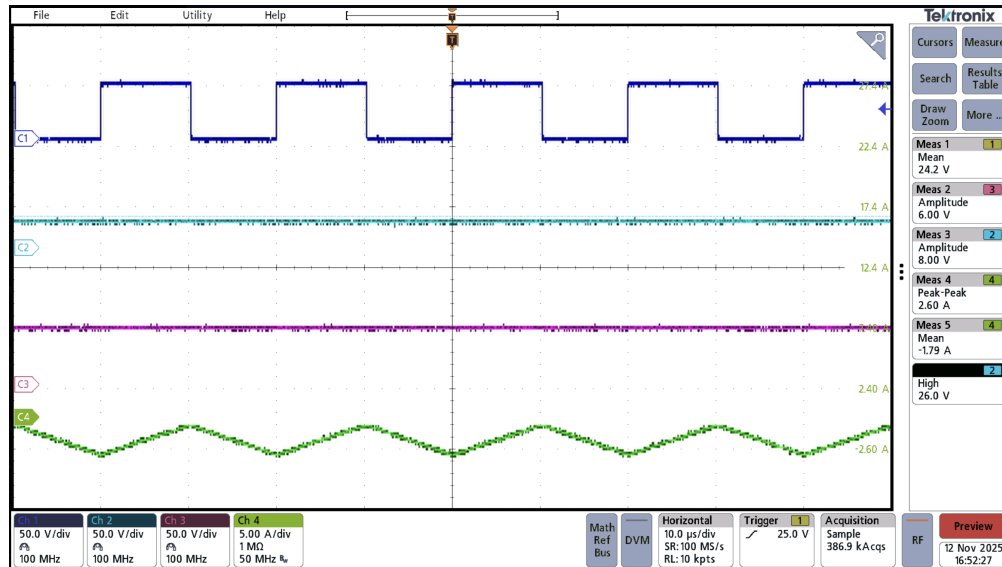


Figure 8: Output Voltage,  $V_o$ , and Input Voltage,  $V_g$ , with AC Coupled Measurements to Show Their Ripple

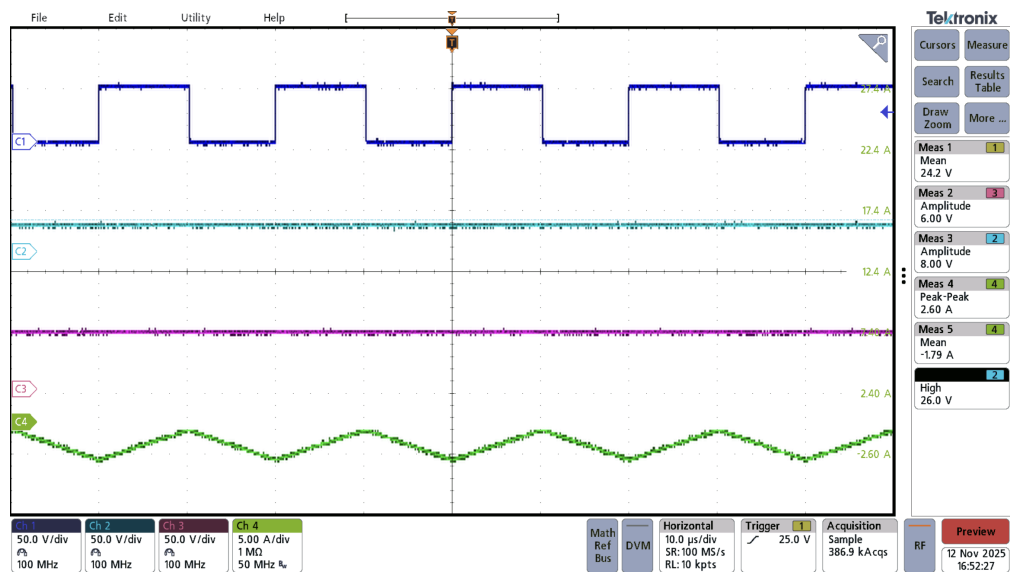


Figure 9:  $V_{sw}$ ,  $V_g$ ,  $V_o$ , and IL waveforms

## 4 ADC Calculations

### 4.1 Interpretation

Higher sampling rates produced more accurate ADC voltage measurements.

**What we learned:** sampling rate directly affects measurement accuracy. **Struggles:** configuring EPwm1Regs registers and making ADC display correct values.

### 4.2 ADC Conversion Code

Listing 3: Voltage back-calculation

```
1 expected_voltage = (AdcaResultRegs.ADCRESULT0 / 4095.0) * 3.3;  
  
2 expected_V_out = (((AdcaResultRegs.ADCRESULT1/4095.0) * 3.3) /  
3 3.149) * 48;  
4 expected_V_g = (((AdcaResultRegs.ADCRESULT2/4095.0) * 3.3) /  
5 3.13) * 24;  
6  
7 expected_I_L = (((AdcaResultRegs.ADCRESULT3/4095.0) * 3.3) -  
8 0.65) * 10;
```

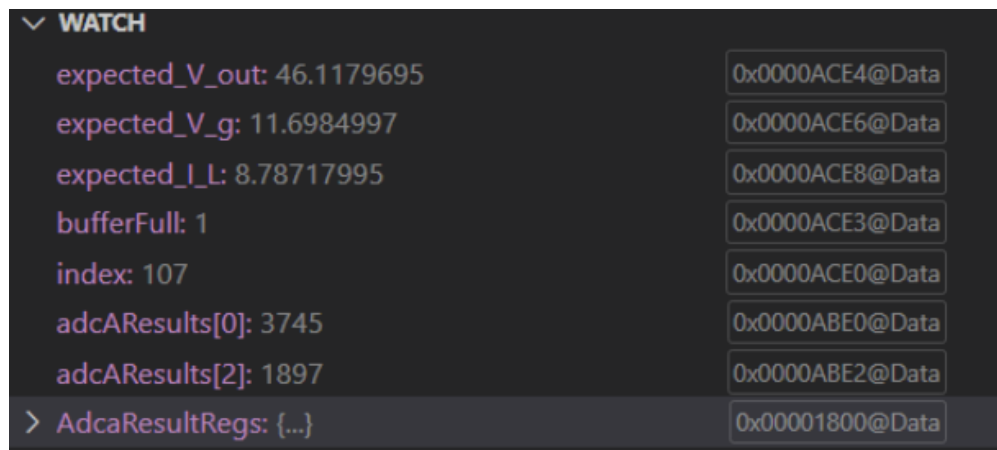


Figure 10: ADC values

## 5 Efficiency

### 5.1 Efficiencies with different loads

(Add your table/plots here)

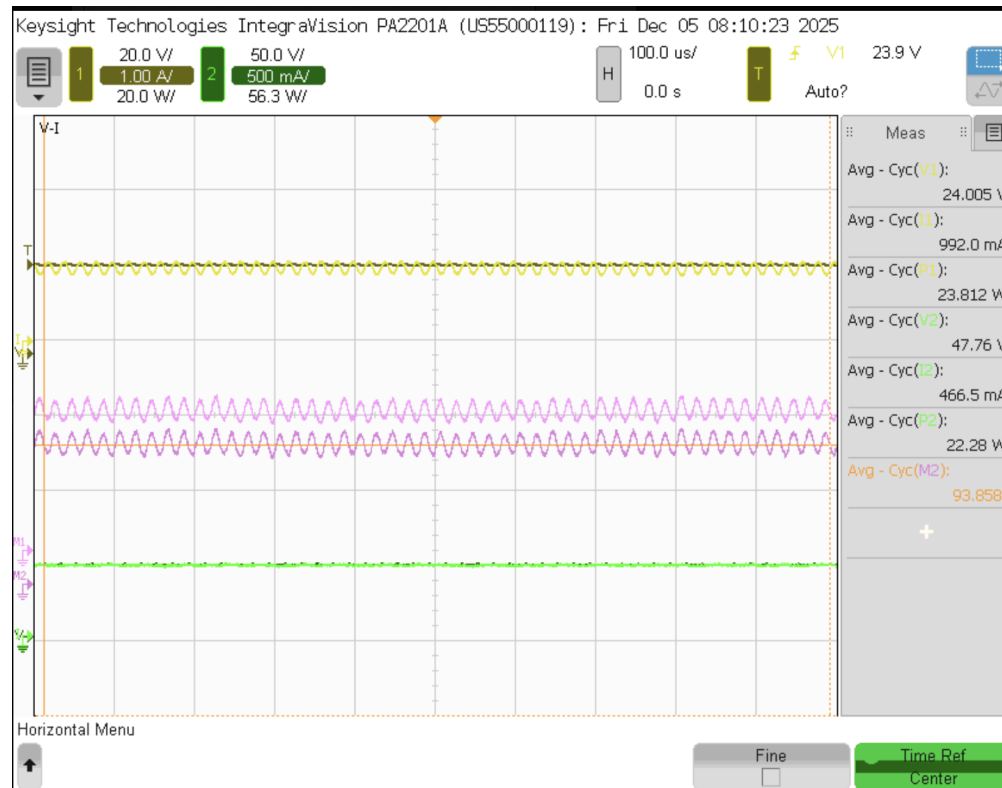


Figure 11: Load at 10%

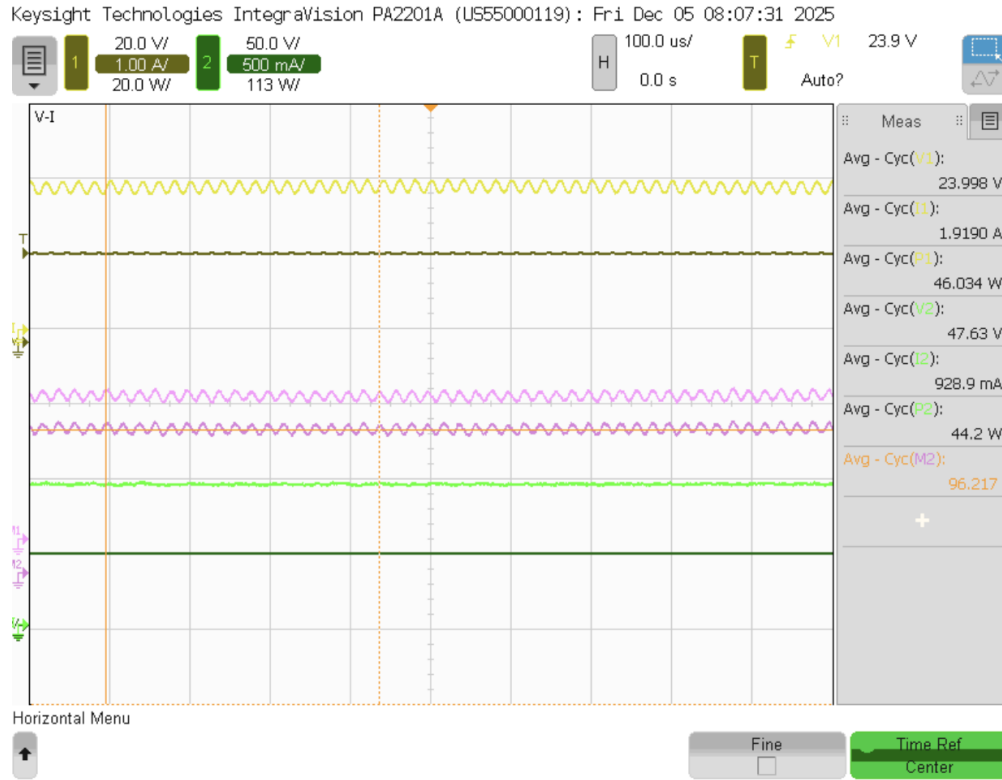


Figure 12: Load at 18%



Figure 13: Load at 37%

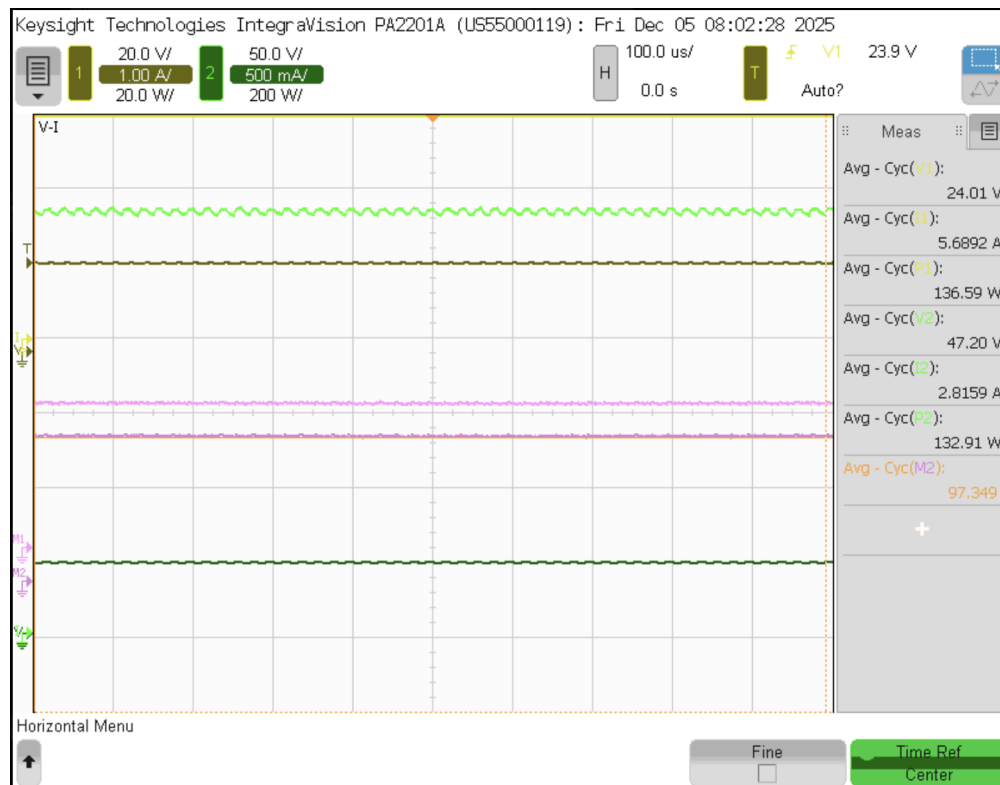


Figure 14: Load at 55%

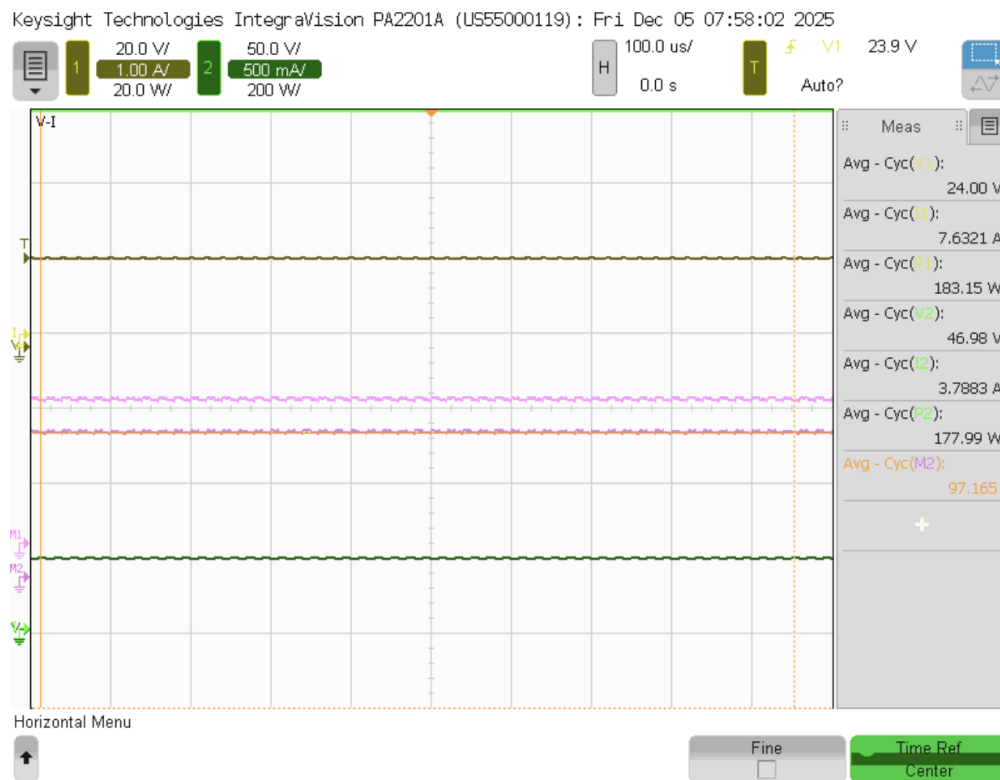


Figure 15: Load at 74%

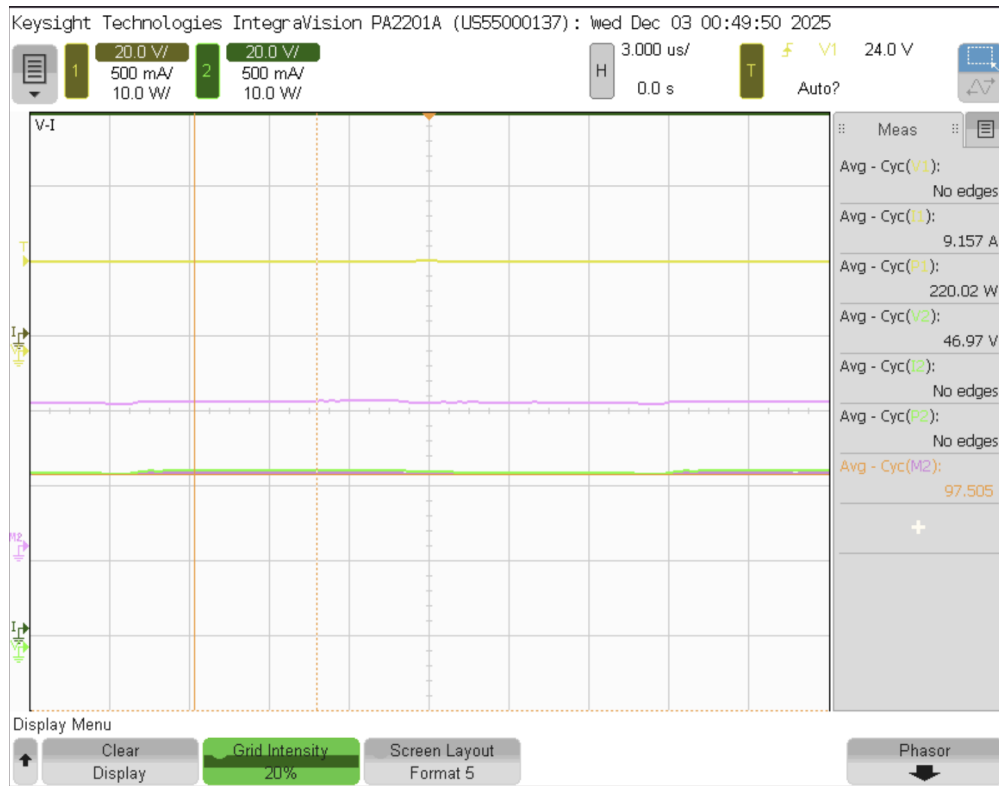


Figure 16: Load at 92% and Max Efficiency

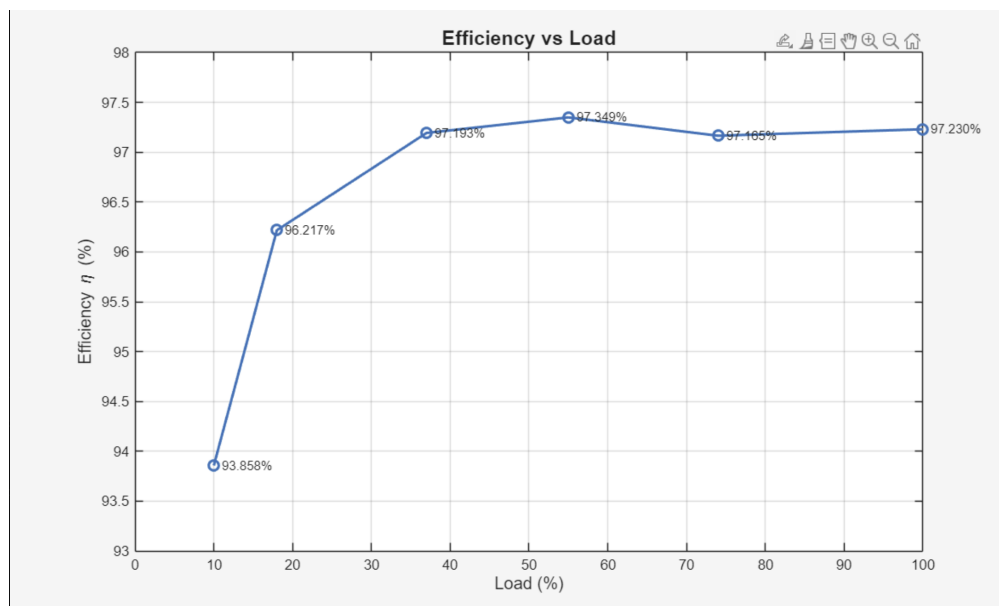


Figure 17: MATLAB Plot for Efficiency

Listing 4: MATLAB Code to Plot Efficiency

```

1  % Load percentages
2  load_pct = [10, 18, 37, 55, 74, 100];
3
4  % Corresponding efficiencies (%)
5  eta = [93.858, 96.217, 97.193, 97.349, 97.165, 97.23];
6
7  % Create figure
8  figure;
9  plot(load_pct, eta, '-o', 'LineWidth', 1.8, 'MarkerSize', 7);
10
11 % Labels and title
12 xlabel('Load (%)', 'FontSize', 12);
13 ylabel('Efficiency\eta (%)', 'FontSize', 12);
14 title('Efficiency vs Load', 'FontSize', 14);
15
16 % Grid and axis formatting
17 grid on;
18 xlim([0 100]);
19 ylim([93 98]);
20
21 % Optional: annotate points
22 for i = 1:length(load_pct)
23     text(load_pct(i) + 1, eta(i), sprintf('%.3f%%', eta(i)), ...
24         'FontSize', 9);
25 end

```

## 6 Conclusion

The boost converter design project taught us so much about design, hardware, and electrical engineering as a whole. In fact, we learned so much throughout this process that it was difficult to condense it down for this lab report. Just in terms of machinery, we learned to work with a power analyzer, LCR meter, and current sensor, since we never operated that equipment before. Although we all have used oscilloscopes before, this project introduced us to the more advanced oscilloscopes, and we learned about differential probes, AC coupling mode, and saving screenshots and files. We also gained experience with new software tools that some of us have never used before, such as Matlab and KiCad. Even for those of us who used KiCad in the past, this project was the first time we worked with pours and had multiple layers on the PCB. We learned about important PCB design considerations that we never really had to make before, such as sensitive signals, high  $di/dt$  and  $dv/dt$  paths, and loop impedance. Furthermore, most of us have coded before, but in the past, we were given values for each register. In this class, we learned how to look through data sheets to find useful information. This also ties in to component selection, since we learned how to use scripts to automate optimization and a figure of merit for a MOSFET. Another new process we came across was inductor customization using the Kg method. It was really cool to see just how many parameters we could adjust to craft the best inductor for our purpose. Overall, this project not only taught us what a boost converter is, but allowed us to fully understand the inner operation of the circuit, each step of the process, and how to balance tradeoffs for an efficient design.

Throughout the semester, there were both highs and lows in terms of what we enjoyed, what we struggled with, and what we struggled with but still really enjoyed. All of us really enjoyed the actual PCB design work. There was so much opportunity to get creative, and we really enjoyed the challenge of cramming the critical and gating loops to be as small as possible. Routing especially is very satisfying. We found the design review presentations really helpful and enjoyed making and presenting our slides. Furthermore, when it came to optimization, we enjoyed getting creative with the different kind of Matlab plots we created to show exactly why our MOSFET was best suited for efficiency. During board assembly, we had a lot of fun soldering SMD components with the hot air gun, as that was also really satisfying. Another privilege we enjoyed was the kind help and encouragement from the TAs as we were working on debugging. Finally, we enjoyed that the full class was framed as a competition, since that gave us a clear goal and provided so much motivation to keep pushing ourselves to achieve the highest efficiency.

On the other hand, there were parts of the process that we struggled with. The main struggle was testing and debugging. The first time we did gate driver testing the converter did not work. We used a multimeter continuity test and figured out that 3.3V was shorted to ground. After desoldering multiple SMD components with no improvement, we were suggested to use a thermal camera and run current through the circuit to see which component actually caused the problem, but we were too scared to do that so we powered through and figured out that our current sensor was flipped. Almost two weeks later, after our board successfully worked, we were testing multiple times under max load for efficiency measurements when our board again, suddenly stopped working. This time, we again desoldered multiple components before

realizing that the MOSFET was internally shorted somehow. Furthermore, another struggle we had was tuning  $R_{small}$ . Despite understanding the theory of how the perfect resistance could lead to a critically damped waveform, when we switched our resistor from the initial 0.5 Ohms to 0 Ohms, there were less oscillations on our waveform, but the converter was still less efficient. Even increasing the resistance didn't work, so we ended up sticking with 0.5 Ohms. Finally, it was also challenging to get our ADC working since one incorrect register value while switching the pins caused many hours of debugging. It was also struggle to get our converter as efficient as it is, but that was a struggle we really enjoyed the process of.

There were also a couple aspects of lab that we disliked and felt could be improved on. The first one was the lack of multiple wire gauge sizes. We calculated a wire gauge of 10 for the most efficient converter, but since that was unavailable, we had to use 14 instead. If that was available, we might've been able to achieve an even more efficient converter. We did feel like the timeline was pretty good, since we finished our converter even with some hiccups along the way and still had time for extra testing, but there was just a lot of extra time commitment that we had to put outside of lab. We were happy to do so, but if it had been a busier week, we might have not been able to achieve the results we did. In addition, for this project we were given a mostly completed schematic. While having this did save a lot of time, we felt that this led to a lack of understanding of all the extra components on the board, such as the protection circuits, certain output capacitors, and current sensor signals. Furthermore, we felt that we didn't fully understand the purpose of every register we modified throughout the labs. One cause of failure and struggle we noticed was that some of the instructions in the lab doc were not updated. For example, instructions about PWM and the instruction to slowly ramp up voltage instead of testing directly at 24 Volts were important considerations that were communicated perhaps later than necessary. We are glad that we switched to hot air soldering because even with experience, soldering those SMD components by hand caused so much frustration because the solder masks were not sufficiently large. Finally, while this relates more to the homework rather than lab, we felt that the course could have been improved by providing instruction, such as a tutorial or easier assignments to start off with for a better introduction to PLECS rather than jumping up to the level of the homework directly.

If we were to redo this lab again, knowing what we do now, there are a couple of things we would do differently. First of all, we would have double checked our soldering more thoroughly to avoid mistakes such as flipping our current sensor. Furthermore, while debugging, we would change the order in which we desolder components, making sure to first test the components that more prone to failure (such as the MOSFETs) before regular SMD components. We would have also used a lower gauge for a bigger inductor wire to result in a higher efficiency. During our testing and data collection, we sent all of our screenshots to each other on discord. If we were to redo this lab, we would catalog our data better by uploading labeled screenshots and our code into a shared drive or folder for easier and more organized access.

Overall, we had a great time designing, building, and testing our boost converter, and this has definitely been our favorite hardware project to date. We are proud of not only our efficient converter, but also of how much we learned along the way. This course truly was

the best hits of ECE all in one and more.